

## SECTION 1: IDENTIFICATION OF THE SUBSTANCE/MIXTURE AND OF THE COMPANY/UNDERTAKING

### 1.1. Product identifier

|                      |                                                       |
|----------------------|-------------------------------------------------------|
| Product Description: | <b>Ethanediol</b>                                     |
| Product Grade:       | SQ, ER                                                |
| Cat No. :            | <b>Q12775, Q12777, Q23405, Q23406, Q23407, Q2340C</b> |
| Synonyms             | Monoethylene glycol; 1,2-Ethanediol                   |
| Index No             | 603-027-00-1                                          |
| CAS No               | 107-21-1                                              |
| EC No                | 203-473-3                                             |
| Molecular Formula    | C2 H6 O2                                              |

### 1.2. Relevant identified uses of the substance or mixture and uses advised against

|                                |                                                                                             |
|--------------------------------|---------------------------------------------------------------------------------------------|
| Recommended Use                | Laboratory chemicals.                                                                       |
| Sector of use                  | SU3 - Industrial uses: Uses of substances as such or in preparations at industrial sites    |
| Product category               | PC21 - Laboratory chemicals                                                                 |
| Process categories             | PROC15 - Use as a laboratory reagent                                                        |
| Environmental release category | ERC6a - Industrial use resulting in manufacture of another substance (use of intermediates) |
| Uses advised against           | No Information available                                                                    |

### 1.3. Details of the supplier of the safety data sheet

|                |                                                                                                                                   |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------|
| Company        | Thermo Fisher Scientific India Pvt. Ltd<br>403-404, B-wing, Delphi,<br>Hiranandani Business Park,<br>Powai, Mumbai 4000076, INDIA |
| E-mail address | laboratorysolution@thermofisher.com                                                                                               |

### 1.4. Emergency telephone number

India Toll Free: 18 00 22 22 30  
Chemtec US: (800) 424-9300  
Chemtec EU: 001 (202) 483-7616

## SECTION 2: HAZARDS IDENTIFICATION

### 2.1. Classification of the substance or mixture

**GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567**

#### **Physical hazards**

Based on available data, the classification criteria are not met

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## Health hazards

Acute oral toxicity  
Specific target organ toxicity - (repeated exposure)

Category 4 (H302)  
Category 2 (H373)

## Environmental hazards

Based on available data, the classification criteria are not met

Full text of Hazard Statements: see section 16

## 2.2. Label elements



Signal Word

Warning

## Hazard Statements

H302 - Harmful if swallowed  
H373 - May cause damage to organs through prolonged or repeated exposure

## Precautionary Statements

P260 - Do not breathe dust/fume/gas/mist/vapors/spray  
P264 - Wash face, hands and any exposed skin thoroughly after handling  
P301 + P330 + P331 - IF SWALLOWED: rinse mouth. Do NOT induce vomiting  
P312 - Call a POISON CENTER or doctor if you feel unwell

## 2.3. Other hazards

Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB)  
Toxic to terrestrial vertebrates  
This product does not contain any known or suspected endocrine disruptors

## SECTION 3: COMPOSITION/INFORMATION ON INGREDIENTS

### 3.1. Substances

| Component       | CAS No   | EC No             | Weight % | GHS Classification - According to GB-CLP Regulations UK SI 2019/720 and UK SI 2020/1567 |
|-----------------|----------|-------------------|----------|-----------------------------------------------------------------------------------------|
| Ethylene glycol | 107-21-1 | EEC No. 203-473-3 | <=100    | Acute Tox. 4 (H302)<br>STOT RE 2 (H373)                                                 |

Full text of Hazard Statements: see section 16

## SECTION 4: FIRST AID MEASURES

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## 4.1. Description of first aid measures

|                                           |                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                     | If symptoms persist, call a physician.                                                                                                           |
| <b>Eye Contact</b>                        | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                  |
| <b>Skin Contact</b>                       | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                |
| <b>Ingestion</b>                          | Clean mouth with water and drink afterwards plenty of water.                                                                                     |
| <b>Inhalation</b>                         | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                     |
| <b>Self-Protection of the First Aider</b> | Ensure that medical personnel are aware of the material(s) involved, take precautions to protect themselves and prevent spread of contamination. |

## 4.2. Most important symptoms and effects, both acute and delayed

None reasonably foreseeable.

## 4.3. Indication of any immediate medical attention and special treatment needed

**Notes to Physician** Treat symptomatically. Symptoms may be delayed.

## SECTION 5: FIREFIGHTING MEASURES

### 5.1. Extinguishing media

#### **Suitable Extinguishing Media**

Water spray, carbon dioxide (CO<sub>2</sub>), dry chemical, alcohol-resistant foam.

#### **Extinguishing media which must not be used for safety reasons**

No information available.

### 5.2. Special hazards arising from the substance or mixture

Thermal decomposition can lead to release of irritating gases and vapors. Keep product and empty container away from heat and sources of ignition.

#### **Hazardous Combustion Products**

Carbon monoxide (CO), Carbon dioxide (CO<sub>2</sub>).

### 5.3. Advice for firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

## SECTION 6: ACCIDENTAL RELEASE MEASURES

### 6.1. Personal precautions, protective equipment and emergency procedures

Ensure adequate ventilation. Use personal protective equipment as required.

### 6.2. Environmental precautions

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Should not be released into the environment. See Section 12 for additional Ecological Information.

## 6.3. Methods and material for containment and cleaning up

Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 6.4. Reference to other sections

Refer to protective measures listed in Sections 8 and 13.

## SECTION 7: HANDLING AND STORAGE

### 7.1. Precautions for safe handling

Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

### 7.2. Conditions for safe storage, including any incompatibilities

Keep containers tightly closed in a dry, cool and well-ventilated place. Keep away from heat, sparks and flame. Store under an inert atmosphere. Protect from moisture.

Technical Rules for Hazardous Substances (TRGS) 510 Class 10  
Storage Class (LGK) (Germany)

### 7.3. Specific end use(s)

Use in laboratories

## SECTION 8: EXPOSURE CONTROLS/PERSONAL PROTECTION

### 8.1. Control parameters

#### Exposure limits

List source(s): **EU** - Commission Directive (EU) 2019/1831 of 24 October 2019 establishing a fifth list of indicative occupational exposure limit values pursuant to Council Directive 98/24/EC and amending Commission Directive 2000/39/EC **UK** - EH40/2005 Work Exposure Limits, Fourth edition. Published 2020. **IRE** - 2021 Code of Practice for the Chemical Agents Regulations, Schedule 1. Published by the Health and Safety Authority

| Component       | The United Kingdom                                                                                                                                                                             | European Union                                                                                                            | Ireland                                                                                                                   |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| Ethylene glycol | STEL: 40 ppm 15 min<br>STEL: 104 mg/m <sup>3</sup> 15 min<br>STEL: 30 mg/m <sup>3</sup> 15 min<br>TWA: 10 mg/m <sup>3</sup> 8 hr<br>TWA: 20 ppm 8 hr<br>TWA: 52 mg/m <sup>3</sup> 8 hr<br>Skin | TWA: 20 ppm (8h)<br>TWA: 52 mg/m <sup>3</sup> (8h)<br>STEL: 40 ppm (15min)<br>STEL: 104 mg/m <sup>3</sup> (15min)<br>Skin | TWA: 20 ppm 8 hr.<br>TWA: 52 mg/m <sup>3</sup> 8 hr.<br>STEL: 40 ppm 15 min<br>STEL: 104 mg/m <sup>3</sup> 15 min<br>Skin |

#### Biological limit values

This product, as supplied, does not contain any hazardous materials with biological limits established by the region specific regulatory bodies

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## Derived No Effect Level (DNEL) / Derived Minimum Effect Level (DMEL)

See table for values

| Component                            | Acute effects local (Dermal) | Acute effects systemic (Dermal) | Chronic effects local (Dermal) | Chronic effects systemic (Dermal) |
|--------------------------------------|------------------------------|---------------------------------|--------------------------------|-----------------------------------|
| Ethylene glycol<br>107-21-1 ( ≤100 ) |                              |                                 |                                | DNEL = 106mg/kg<br>bw/day         |

| Component                            | Acute effects local (Inhalation) | Acute effects systemic (Inhalation) | Chronic effects local (Inhalation)                         | Chronic effects systemic (Inhalation) |
|--------------------------------------|----------------------------------|-------------------------------------|------------------------------------------------------------|---------------------------------------|
| Ethylene glycol<br>107-21-1 ( ≤100 ) |                                  |                                     | DNEL = 35mg/m <sup>3</sup><br>DNEL = 33.5mg/m <sup>3</sup> | DNEL = 70mg/m <sup>3</sup>            |

## Predicted No Effect Concentration (PNEC)

See values below.

| Component                            | Fresh water                      | Fresh water sediment                                                                              | Water Intermittent              | Microorganisms in sewage treatment | Soil (Agriculture)                                                                        |
|--------------------------------------|----------------------------------|---------------------------------------------------------------------------------------------------|---------------------------------|------------------------------------|-------------------------------------------------------------------------------------------|
| Ethylene glycol<br>107-21-1 ( ≤100 ) | PNEC = 10mg/L<br>PNEC = 85.9mg/L | PNEC = 37mg/kg<br>sediment dw<br>PNEC = 312mg/kg<br>sediment dw<br>PNEC = 317mg/kg<br>sediment dw | PNEC = 10mg/L<br>PNEC = 130mg/L | PNEC = 199.5mg/L<br>PNEC = 200mg/L | PNEC = 1.53mg/kg<br>soil dw<br>PNEC = 12.7mg/kg<br>soil dw<br>PNEC = 13.1mg/kg<br>soil dw |

| Component                            | Marine water                    | Marine water sediment                                                                                | Marine water intermittent | Food chain | Air |
|--------------------------------------|---------------------------------|------------------------------------------------------------------------------------------------------|---------------------------|------------|-----|
| Ethylene glycol<br>107-21-1 ( ≤100 ) | PNEC = 1mg/L<br>PNEC = 8.59mg/L | PNEC = 3.7mg/kg<br>sediment dw<br>PNEC = 31.2mg/kg<br>sediment dw<br>PNEC = 31.7mg/kg<br>sediment dw | PNEC = 10mg/L             |            |     |

## 8.2. Exposure controls

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

#### Eye Protection

Wear safety glasses with side shields (or goggles) (European standard - EN 166)

#### Hand Protection

Protective gloves

| Glove material | Breakthrough time                 | Glove thickness | EU standard | Glove comments        |
|----------------|-----------------------------------|-----------------|-------------|-----------------------|
| Viton (R)      | See manufacturers recommendations | -               | EN 374      | (minimum requirement) |

#### Skin and body protection

Long sleeved clothing.

Inspect gloves before use.

Please observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves.

(Refer to manufacturer/supplier for information)

Ensure gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g.

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sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion.

Remove gloves with care avoiding skin contamination.

## Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators.  
To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

## Large scale/emergency use

Use a NIOSH/MSHA or European Standard EN 136 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced  
**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

## Small scale/Laboratory use

Use a NIOSH/MSHA or European Standard EN 149:2001 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

**Recommended half mask:-** Valve filtering: EN405; or; Half mask: EN140; plus filter, EN 141

When RPE is used a face piece Fit Test should be conducted

## Environmental exposure controls

No information available.

## SECTION 9: PHYSICAL AND CHEMICAL PROPERTIES

### 9.1. Information on basic physical and chemical properties

|                                         |                                                 |  |                           |
|-----------------------------------------|-------------------------------------------------|--|---------------------------|
| Physical State                          | Viscous liquid Liquid                           |  |                           |
| Appearance                              | Colorless                                       |  |                           |
| Odor                                    | Odorless                                        |  |                           |
| Odor Threshold                          | No data available                               |  |                           |
| Melting Point/Range                     | -13 °C / 8.6 °F                                 |  |                           |
| Softening Point                         | No data available                               |  |                           |
| Boiling Point/Range                     | 196 - 198 °C / 384.8 - 388.4 °F @ 760 mmHg      |  |                           |
| Flammability (liquid)                   | No data available                               |  |                           |
| Flammability (solid,gas)                | Not applicable                                  |  | Liquid                    |
| Explosion Limits                        | <b>Lower</b> 3.2 vol %<br><b>Upper</b> 28 vol % |  |                           |
| Flash Point                             | 111 °C / 231.8 °F                               |  | <b>Method -</b> DIN 51758 |
| Autoignition Temperature                | 413 °C / 775.4 °F                               |  |                           |
| Decomposition Temperature               | > 500°C                                         |  |                           |
| pH                                      | 5.5-7.5                                         |  | 50% aq. sol               |
| Viscosity                               | 21 cP (20°C)                                    |  |                           |
| Water Solubility                        | Miscible                                        |  |                           |
| Solubility in other solvents            | No information available                        |  |                           |
| Partition Coefficient (n-octanol/water) |                                                 |  |                           |
| Component                               | <b>log Pow</b>                                  |  |                           |
| Ethylene glycol                         | -1.36                                           |  |                           |
| Vapor Pressure                          | 0.12 mmHg @ 20 °C                               |  |                           |
| Density / Specific Gravity              | 1.113                                           |  |                           |
| Bulk Density                            | Not applicable                                  |  | Liquid                    |
| Vapor Density                           | 2.14 (Air = 1.0)                                |  | (Air = 1.0)               |
| Particle characteristics                | Not applicable (liquid)                         |  |                           |

### 9.2. Other information

|                   |                          |
|-------------------|--------------------------|
| Molecular Formula | C2 H6 O2                 |
| Molecular Weight  | 62.06                    |
| Evaporation Rate  | No information available |

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## SECTION 10: STABILITY AND REACTIVITY

### 10.1. Reactivity

None known, based on information available

### 10.2. Chemical stability

Hygroscopic.

### 10.3. Possibility of hazardous reactions

#### Hazardous Polymerization Hazardous Reactions

Hazardous polymerization does not occur.  
None under normal processing.

### 10.4. Conditions to avoid

Incompatible products. Excess heat. Exposure to moist air or water.

### 10.5. Incompatible materials

Strong oxidizing agents. Strong acids. Strong bases. Aldehydes.

### 10.6. Hazardous decomposition products

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>).

## SECTION 11: TOXICOLOGICAL INFORMATION

### 11.1. Information on hazard classes as defined in Regulation (EC) No 1272/2008

#### Product Information

#### (a) acute toxicity;

Oral

Category 4

Dermal

Based on available data, the classification criteria are not met

Inhalation

Based on available data, the classification criteria are not met

| Component       | LD50 Oral                                                       | LD50 Dermal                                                                            | LC50 Inhalation             |
|-----------------|-----------------------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------------|
| Ethylene glycol | 7712 mg/kg ( Rat )<br>Human LD50 estimate:<br>1,400–1,600 mg/kg | LD50 = 9530 µL/kg ( Rabbit )<br>LD50 = 10600 mg/kg ( Rat )<br>LD50 > 3500 mg/kg (mice) | LC50 > 2.5 mg/L ( Rat ) 6 h |

#### (b) skin corrosion/irritation;

Based on available data, the classification criteria are not met

#### (c) serious eye damage/irritation;

Based on available data, the classification criteria are not met

#### (d) respiratory or skin sensitization;

Respiratory

Based on available data, the classification criteria are not met

Skin

Based on available data, the classification criteria are not met

#### (e) germ cell mutagenicity;

Based on available data, the classification criteria are not met

#### (f) carcinogenicity;

Based on available data, the classification criteria are not met

There are no known carcinogenic chemicals in this product

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(g) reproductive toxicity; Based on available data, the classification criteria are not met

(h) STOT-single exposure; Based on available data, the classification criteria are not met

(i) STOT-repeated exposure; Category 2

Target Organs Central nervous system (CNS), Liver, Kidney.

(j) aspiration hazard; Based on available data, the classification criteria are not met

Symptoms / effects, both acute and delayed No information available.

## 11.2. Information on other hazards

Endocrine Disrupting Properties Assess endocrine disrupting properties for human health. This product does not contain any known or suspected endocrine disruptors.

## SECTION 12: ECOLOGICAL INFORMATION

### 12.1. Toxicity

Ecotoxicity effects Do not empty into drains. .

| Component       | Freshwater Fish                                                                                                                                                                                                                                                                                                                             | Water Flea                              | Freshwater Algae                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------|----------------------------------------------------------------|
| Ethylene glycol | LC50: = 41000 mg/L, 96h (Oncorhynchus mykiss)<br>LC50: = 27540 mg/L, 96h static (Lepomis macrochirus)<br>LC50: 14 - 18 mL/L, 96h static (Oncorhynchus mykiss)<br>LC50: = 40761 mg/L, 96h static (Oncorhynchus mykiss)<br>LC50: 40000 - 60000 mg/L, 96h static (Pimephales promelas)<br>LC50: = 16000 mg/L, 96h static (Poecilia reticulata) | EC50: = 46300 mg/L, 48h (Daphnia magna) | EC50: 6500 - 13000 mg/L, 96h (Pseudokirchneriella subcapitata) |

12.2. Persistence and degradability Readily biodegradable  
Persistence Persistence is unlikely.

12.3. Bioaccumulative potential Bioaccumulation is unlikely

| Component       | log Pow | Bioconcentration factor (BCF) |
|-----------------|---------|-------------------------------|
| Ethylene glycol | -1.36   | No data available             |

12.4. Mobility in soil The product is water soluble, and may spread in water systems Will likely be mobile in the environment due to its water solubility. Highly mobile in soils

12.5. Results of PBT and vPvB assessment Substance is not considered persistent, bioaccumulative and toxic (PBT) / very persistent and very bioaccumulative (vPvB).

### 12.6. Endocrine disrupting

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## properties

**Endocrine Disruptor Information** This product does not contain any known or suspected endocrine disruptors

## 12.7. Other adverse effects

**Persistent Organic Pollutant** This product does not contain any known or suspected substance

**Ozone Depletion Potential** This product does not contain any known or suspected substance

## SECTION 13: DISPOSAL CONSIDERATIONS

### 13.1. Waste treatment methods

**Waste from Residues/Unused Products** Waste is classified as hazardous. Dispose of in accordance with the European Directives on waste and hazardous waste. Dispose of in accordance with local regulations.

**Contaminated Packaging** Dispose of this container to hazardous or special waste collection point.

**European Waste Catalogue (EWC)** According to the European Waste Catalog, Waste Codes are not product specific, but application specific.

**Other Information** Waste codes should be assigned by the user based on the application for which the product was used. Do not empty into drains.

## SECTION 14: TRANSPORT INFORMATION

**IMDG/IMO** Not regulated

### 14.1. UN number

### 14.2. UN proper shipping name

### 14.3. Transport hazard class(es)

### 14.4. Packing group

**ADR** Not regulated

### 14.1. UN number

### 14.2. UN proper shipping name

### 14.3. Transport hazard class(es)

### 14.4. Packing group

**IATA** Not regulated

### 14.1. UN number

### 14.2. UN proper shipping name

### 14.3. Transport hazard class(es)

### 14.4. Packing group

**14.5. Environmental hazards** No hazards identified

**14.6. Special precautions for user** No special precautions required.

**14.7. Maritime transport in bulk according to IMO instruments** Not applicable, packaged goods

## SECTION 15: REGULATORY INFORMATION

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## 15.1. Safety, health and environmental regulations/legislation specific for the substance or mixture

### International Inventories

Europe (EINECS/ELINCS/NLP), China (IECSC), Taiwan (TCSI), Korea (KECL), Japan (ENCS), Japan (ISHL), Canada (DSL/NDSL), Australia (AICS), New Zealand (NZIoC), Philippines (PICCS). US EPA (TSCA) - Toxic Substances Control Act, (40 CFR Part 710)

| Component       | CAS No   | EINECS    | ELINCS | NLP | IECSC | TCSI | KECL     | ENCS | ISHL |
|-----------------|----------|-----------|--------|-----|-------|------|----------|------|------|
| Ethylene glycol | 107-21-1 | 203-473-3 | -      | -   | X     | X    | KE-13169 | X    | X    |

| Component       | CAS No   | TSCA | TSCA Inventory notification - Active-Inactive | DSL | NDSL | AICS | NZIoC | PICCS |
|-----------------|----------|------|-----------------------------------------------|-----|------|------|-------|-------|
| Ethylene glycol | 107-21-1 | X    | ACTIVE                                        | X   | -    | X    | X     | X     |

Legend: X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

### Authorisation/Restrictions according to EU REACH

Not applicable

| Component       | CAS No   | REACH (1907/2006) - Annex XIV - Substances Subject to Authorization | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous Substances | REACH Regulation (EC 1907/2006) article 59 - Candidate List of Substances of Very High Concern (SVHC) |
|-----------------|----------|---------------------------------------------------------------------|-------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| Ethylene glycol | 107-21-1 | -                                                                   | -                                                                             | -                                                                                                     |

### Seveso III Directive (2012/18/EC)

| Component       | CAS No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements |
|-----------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Ethylene glycol | 107-21-1 | Not applicable                                                                            | Not applicable                                                                           |

### Regulation (EC) No 649/2012 of the European Parliament and of the Council of 4 July 2012 concerning the export and import of dangerous chemicals

Not applicable

### Contains component(s) that meet a 'definition' of per & poly fluoroalkyl substance (PFAS)?

Not applicable

Take note of Directive 98/24/EC on the protection of the health and safety of workers from the risks related to chemical agents at work .

Take note of Directive 2000/39/EC establishing a first list of indicative occupational exposure limit values

### National Regulations

UK - Take note of Control of Substances Hazardous to Health Regulations (COSHH) 2002 and 2005 Amendment

### WGK Classification

See table for values

| Component       | Germany - Water Classification (AwSV) | Germany - TA-Luft Class |
|-----------------|---------------------------------------|-------------------------|
| Ethylene glycol | WGK1                                  |                         |

| Component       | France - INRS (Tables of occupational diseases)      |
|-----------------|------------------------------------------------------|
| Ethylene glycol | Tableaux des maladies professionnelles (TMP) - RG 84 |

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| Component                             | Switzerland - Ordinance on the Reduction of Risk from handling of hazardous substances preparation (SR 814.81) | Switzerland - Ordinance on Incentive Taxes on Volatile Organic Compounds (OVOC) | Switzerland - Ordinance of the Rotterdam Convention on the Prior Informed Consent Procedure |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|
| Ethylene glycol<br>107-21-1 ( <=100 ) | Prohibited and Restricted Substances                                                                           |                                                                                 |                                                                                             |

## 15.2. Chemical safety assessment

A Chemical Safety Assessment/Report (CSA/CSR) has been conducted by the manufacturer/importer

## SECTION 16: OTHER INFORMATION

### Full text of H-Statements referred to under sections 2 and 3

H302 - Harmful if swallowed

H373 - May cause damage to organs through prolonged or repeated exposure

### Legend

**CAS** - Chemical Abstracts Service

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**DSL/NDL** - Canadian Domestic Substances List/Non-Domestic Substances List

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**NZIoC** - New Zealand Inventory of Chemicals

**WEL** - Workplace Exposure Limit

**ACGIH** - American Conference of Governmental Industrial Hygienists

**DNEL** - Derived No Effect Level

**RPE** - Respiratory Protective Equipment

**LC50** - Lethal Concentration 50%

**NOEC** - No Observed Effect Concentration

**PBT** - Persistent, Bioaccumulative, Toxic

**TWA** - Time Weighted Average

**IARC** - International Agency for Research on Cancer  
Predicted No Effect Concentration (PNEC)

**LD50** - Lethal Dose 50%

**EC50** - Effective Concentration 50%

**POW** - Partition coefficient Octanol:Water

**vPvB** - very Persistent, very Bioaccumulative

**ADR** - European Agreement Concerning the International Carriage of Dangerous Goods by Road

**IMO/IMDG** - International Maritime Organization/International Maritime Dangerous Goods Code

**OECD** - Organisation for Economic Co-operation and Development

**BCF** - Bioconcentration factor

### Key literature references and sources for data

<https://echa.europa.eu/information-on-chemicals>

Suppliers safety data sheet, Chemadvisor - LOLI, Merck index, RTECS

**ICAO/IATA** - International Civil Aviation Organization/International Air Transport Association

**MARPOL** - International Convention for the Prevention of Pollution from Ships

**ATE** - Acute Toxicity Estimate

**VOC** - (Volatile Organic Compound)

### Training Advice

Chemical hazard awareness training, incorporating labelling, Safety Data Sheets (SDS), Personal Protective Equipment (PPE) and hygiene.

Use of personal protective equipment, covering appropriate selection, compatibility, breakthrough thresholds, care, maintenance, fit and standards.

First aid for chemical exposure, including the use of eye wash and safety showers.

Chemical incident response training.

Fire prevention and fighting, identifying hazards and risks, static electricity, explosive atmospheres posed by vapours and dusts.

**Creation Date** 02-Feb-2010

**Revision Date** 07-Nov-2024

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Revision Summary

Next Revision Date - May 2029.

**This safety data sheet complies with Regulation UK SI 2019/758 and UK SI 2020/1577 as amended.**

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of Safety Data Sheet**